INCH-POUND

MIL-C-52921 (ME)
28 December 1976

MILITARY SPECIFICATION
CRUSHING, SCREENING AND WASHING PLANT, 150 TONS PER HOUR, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers a commercial type crushing, screening, and washing plant consisting of mobile crushing and screening units and auxiliary mobile equipment mounted on pneumatic tired wheels compatible for towing by various trucks and truck tractors.

2. APPLICABLE DOCUMENTS.

2. Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-E-52798	Enamel, Alkyd, Camouflage.

STANDARDS

FEDERAL

FED-STD-595	Colors

(Copies of specification and standard should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3820

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C-33 – Concrete Aggregates, Specification for.
D-1073 – Fine Aggregate for Bituminous Paving Mixtures, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

US DEPARTMENT OF LABOR

Occupational Safety and Health Act (OSHA) Standards.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to the US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-H, Fort Belvoir, VA 22060.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA Standards.

(Application for copies should be addressed to the National Electric Manufacturers Association, 155 East 44th Street, New York NY 10017.)

DEPARTMENT OF TRANSPORTATION FEDERAL MOTOR CARRIER SAFETY REGULATIONS.

Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference at libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS.

3.1 Description. The plant shall consist of semitrailer mounted, electric motor driven rock crushing, screening and washing units and auxiliary equipment mounted on pneumatic tired wheels capable of being towed over highways, secondary roads and cross country by commercial truck tractor or military equipment truck tractor. The crushing, screening and washing plant shall meet the applicable requirements of the Department of Transportation (DOT) and the Occupational Safety and Health Act, (OSHA) Standards as specified herein and which are in effect for this type of equipment on the date of the opening of bids. The crushing and screening plant shall be the current commercial production model of a standard product and, except

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for product improvements, optional equipments and accessories required to meet the requirements specified herein, shall have demonstrated industry acceptability by having been manufactured, marketed and sold in significant numbers to the commercial public for at least one year prior to the opening of the technical proposals or bid. In the case of a recently introduced item, the manufacturer will submit sufficient valid industrial operational/test data to demonstrate the acceptability of the item. Each unit of the plant shall be furnished complete with all necessary electric motor power units, hoppers, feeders and conveyors for receiving quarry materials and delivering the finished product to the succeeding unit or to a stock pile. All necessary detail parts shall be included to make the plant complete for the performance of all production functions as specified herein. The plant shall be furnished in two basic configurations, each capable of independent operation, or being combined for operation as a complete plant capable of producing a full range of products as follows:

(a) Construction plant. A construction plant configuration shall consist of a jaw type primary crusher, a secondary crushing and screening unit, and related equipment to produce base stone and concrete aggregate.
(b) Construction support plant. The construction support plant configuration shall consist of all the construction plant components with the addition of a tertiary crushing unit, a washing and screening unit, portable conveyors and other related equipment required to provide a complete and operational crushing, screening and washing plant to produce concrete stone, asphalt aggregate and sand, washed or unwashed, to specification requirements.

3.1.1 Plant. Each plant configuration shall meet the Department of Transportation (DOT) Federal Motor Carrier Safety Regulations for over-the-highway use and the Occupational Safety and Health Act (OSHA) Standards which are in effect for this type of equipment on the date of the opening of the bids.

3.2 Performance. The combined crushing, screening and washing plant shall be capable of receiving materials from a rock quarry or gravel pit and performing the necessary reduction and processing to produce finished aggregate in quality or quantity as specified in 3.3.1 and 3.5.7. The plant shall be capable of dry operation in ambient temperatures from -25° F to 110° F. All production and capacity requirements stated herein shall be based on crushing limestone weighing not less that 2600 per cubic yard (pcy) loose with feed sizes to the primary crusher to be no less than 3 inches smaller than the feed opening of the crusher in at least one dimension. The plant shall also be capable of producing base stone and aggregate from any rock quarry having stone with a compressive strength range from 20,000 to 32,000 psi.

3.2.1 Electric power motors. All electric motors shall be 240/416 volt; 3-phase, 60-cycle (Hz) and shall be sufficient size to maintain a continuous flow of materials through the plant and meet production requirements as specified herein. The motors shall be totally enclosed fan cooled type having a continuous horsepower rating equal to the operating load requirement of the plant components. All electric motors furnished shall comply with NEMA standards for this application.

3.2.2 Product conveyors. Steel channel or lattice frame type product conveyors of sufficient number, lengths, size and capacity to maintain a continuous flow of materials between the crushing and screening units, discharge all products from the units and stockpile all products as required for continuous operation and meet production requirements as specified shall be furnished. All conveyors over 15 feet in length shall be driven from the discharge end through a torque-arm reduction type drive with back-stop to prevent belt rollback. The conveyors shall be equipped with loading hoppers and rubber flashings of sufficient lengths to minimize material spillage. The head or drive pulleys shall be crowned and lagged. The tail pulleys shall be the open or self-cleaning type with a screw type take-up adjustment for belt tensioning. The belt carrier rolls shall be the troughing type. Impact-type rolls shall be provided under the belts at all loading points and under receiving hoppers. Belt wipers shall be provided on each conveyor to prevent material build-up on conveyor belts. The under-carriage of the conveyors shall be provided with hydraulically operated telescoping members to permit raising and lowering the conveyor head as required for operation. The conveyors shall be mounted on pneumatic tired, swivel-type wheels and shall be capable of being towed over highways or cross-country by truck, truck tractor or other towing vehicle.

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3.2.3 Internal conveyors. All internal conveyors as employed for the specific functions within the crushing and for screening units shall be provided with belt carrier rolls, belt flashings, and provisions for belt adjustments as specified in 3.2.2.

3.3 Construction plant.

3.3.1 Primary unit. The primary unit shall consist of a semitrailer chassis upon which shall be mounted a feeder and feed hopper, jaw type crusher, end discharge conveyor, fines reject conveyor, gating mechanisms to control flow materials and electric drive unit.

3.3.1.1 Feeder. A vibrating or reciprocating plate feeder with a multi-deck adjustable bar type grizzly shall be furnished. Materials retained on the grizzly will be discharged into the jaw crusher feed hopper. Materials retained on the lower decks shall be discharged into either the under crusher hopper and mixed with the crushed materials or discharged to a side product conveyor, or a fines reject conveyor, or both with the flow of materials being controlled as desired by gating mechanisms.

3.3.1.2 Reject conveyor. A belt type channel frame reject conveyor shall be furnished. The conveyor shall be capable of removing all or any portion of the fines passing the grizzly. Materials to be discharged by the reject conveyor shall be controlled by an adjustable gating arrangement. The reject conveyor shall be of either removable or the hinged type to reduce width when necessary.

3.3.1.3 Jaw crusher. An overhead eccentric, single toggle type jaw crusher shall be furnished. The crusher shall be provided with a feed hopper for receiving materials and shall discharge crushed materials onto an under crusher belt type discharge conveyor. The crusher shall be capable of crushing all materials ranging in size from a cross section having length and width dimensions 2 inches less than the top opening dimensions of the crusher jaws down to the size of the minimum discharge opening. The discharge opening of the crusher shall be provided with a means for adjustment. The crusher shall be provided with replaceable manganese steel jaws and cheek plates.

3.3.1.4 End delivery conveyor. The primary crushing unit shall be provided with an integral belt type channel frame end delivery conveyor. The conveyor shall be capable of receiving all materials from the grizzly chutes and under crusher hopper and discharging the materials into feed hopper of the succeeding crushing unit or into the receiving hopper of a portable conveyor for transfer to succeeding unit or to a stockpile. The conveyor shall be equipped with impact type belt carrier rolls as specified in 3.2.2, under the crusher discharge hopper at the point where crushed materials impact on the conveyor belt. The conveyor belt shall be driven from the discharge end through a torque-arm type drive. The conveyor shall be either the knock-down or folding type for reducing length for transport purposes.

3.3.2 Secondary unit. A semitrailer mounted electric motor driven, crushing, screening and processing unit capable of receiving materials from the primary crushing unit, removing the fines, crushing oversize products, and blending products to meet aggregate requirements. The secondary unit shall be furnished complete and shall be capable of being operated separately in a gravel pit type operation to produce limited products independent of the other crushing plant units. The unit shall include, but not be limited to, the following components.

3.3.2.1 Feeder. A reciprocating plate or vibratory type feeder with hopper shall be furnished. The hopper shall be mounted on the semitrailer chassis and shall be of sufficient size to receive the materials from the primary crusher and maintain an adequate flow of materials to the overhead feed conveyor to the screening unit. A removable screen or bar type grizzly shall be furnished and mounted over the feed hopper for scalping oversize materials.

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3.3.2.2 Feed conveyor. A channel frame, belt type overhead feed conveyor shall be furnished. The conveyor shall receive materials from the feed hopper and rotary elevator or return conveyor and discharge into hopper over the top deck of the screening unit The belt shall be driven through a torque-arm arrangement from the head end. The conveyor shall be equipped with carrier and return rolls and drive pulleys as specified in 3.2.2.

3.3.2.3 Screening unit. A triple-deck vibrator screening unit equipped with screen trays having sufficient capacity and sized to meet product requirements shall be furnished. The screening unit shall be complete with all hoppers, chutes, and gating mechanisms to receive, screen, blend, and discharge finished products to specification requirements. The screening unit shall be constructed with provisions to permit mounting and utilization of spray bar type washing equipment. The washing equipment when installed, shall not preclude the screening unit from being used for dry type screening operations when aggregate washing is not required.

3.3.2.4 Rotary elevator. A rotary bucket type return elevator with adjustable and replaceable buckets shall be furnished. The elevator shall receive crushed material from the undercrusher conveyor and discharge the material onto the overhead feed conveyor for screening.

3.3.2.5 Under crusher(s) conveyor. A channel frame belt type, under crusher return conveyor shall be furnished. The conveyor shall be equipped with carrier and return rolls and drive pulleys and flashings as specified in 3.2.2. Additional flashings and other provisions to minimize spillage under the crusher shall be provided. The conveyor shall receive materials from the crusher discharge hoppers and discharge the crushed material onto the rotary elevator for recirculation to the screening unit. The conveyor may be either belt driven from the crusher power unit or driven by a separate electric motor.

3.3.2.6 Crushers. The secondary crushing and screening unit shall utilize either roll, cone, or jaw type standard crushers. The crushers shall be provided with automatic protection against overload and damage due to tramp iron or other uncrushable materials that may enter the crushing chamber. The discharge openings of the crushers shall be adjustable to permit production of a full range of aggregate. The crushers shall be provided with under crusher hoppers and chutes which will permit combined type operation to produce materials to various specification requirements. The crusher electric drive motors shall be mounted on platform type adjustable bases to facilitate drive belt adjustment and replacement.

3.3.2.7 Production. The construction plant shall be capable of producing from quarry material, a minimum of 250 tons per hour of product suitable for base stone or a minimum of 180 tons per hour of product suitable for concrete aggregate.

(a) Base stone requirements shall meet the following specifications: AASHO Designation M77 (3-1/2 to 1-1/2 inch size).

Sieve passed	Percent passing
4"	100
3-1/2"	95 — 100
2-1/2"	25 — 60
1-1/2"	0 — 15
3/4"	0 — 5

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(b) Concrete stone shall meet the following specifications: (ASTM C-33, No. 467, 1-1/2" top size to #4).

Sieve passed	Percent passing
2"	100
1-1/2"	95 — 100
3/4"	35 — 70
3/8"	10 — 30
#4	0 — 5

All production rates shall be based on crushing limestone weighing approximately 2600 pounds per cubic yard loose.

3.4 Construction support plant.

3.4.1 General. The construction support will consist of all the components of construction plant with the addition of a screening and washing unit, a screw-type sand classifier, a tertiary crushing unit, conveyors and all accessory equipment as required to provide an increased asphalt aggregate and sand producing capability. All units as required to make up the construction support plant shall be furnished complete with all components, accessories, and hardware, whether or not specified herein, as required to be combined with the construction plant for simultaneous operation as a complete construction support plant; capable of producing and stockpiling concrete aggregate, asphalt aggregate and sand to specification requirements, as stated herein.

3.4.2 Screening and washing unit. A portable, semitrailer mounted, triple deck, vibratory screening, washing and processing unit shall be furnished. The screening unit shall be equipped with water spray bars installed over screen decks for aggregate washing. The screening and washing unit shall receive materials from the construction plant, removed sized materials requiring additional crushing, and screen and blend all products as required to meet specification requirements. The screening and washing unit shall be provided with a receiving hopper under the bottom screen deck and have provisions for adapting to the hopper of screw type sand classifier-dehydrator. The screening and washing unit shall be furnished complete with all wash water plumbing equipment, waste water discharge flumes, product discharge chutes, and blending gates for processing all materials from the washing and screening unit.

3.4.3 Classifier-dehydrator. A twin or single screw sand classifier-dehydrator shall be furnished. The classifier shall receive all fines and water passing the bottom screen deck, wash and remove water from the material and discharge the sand through chutes to product conveyor.

3.4.4 Crusher. A portable semitrailer mounted, roll or cone type single pass tertiary crusher shall be furnished. The crusher shall be capable of closed circuit type operation when combined with the screening and washing unit for production of additional fines and sand when required. The crusher shall receive oversize material from the screening unit, perform further size reduction and discharge the crushed material back to the feed conveyor or hopper of the screening unit. The tertiary crusher shall be furnished complete with all components and accessories.

3.4.5 Production. The construction support plant shall be capable of producing base stone and concrete aggregate at the same capacity as the construction plant with the additional requirements to produce asphalt aggregates (washed or not washed) at a minimum rate of 150 tons per hour. Asphalt aggregate requirements shall meet ASTM C-33, Size No. 507.

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Sieve passed	Percent passing
1-1/2"	100
1"	95 — 100
1/2"	25 — 60
#4	0 — 10
#8	0 — 5

The construction support plant shall also be capable of producing products for use as concrete and asphalt sand. Concrete sand shall meet ASTM C-33 requirements for fine aggregate. Asphalt sand shall meet the requirements for fine aggregate. Asphalt sand shall meet the requirements of ASTM D-1073, Grade No. 1 (3/8" - 0). No minimum sand production capability is specified. However, it is desired that the plant produce at least 50 tons per hour. The contractor shall indicate the calculated sand producing capability of the construction plant offered. All capacities shall be based on crushing and processing limestone weighing approximately 2600 pcy loose with feed sizes as specified in 3.2.

3.4.6 Water pumping unit. A gasoline or diesel engine driven water pump shall be furnished as a water supply source for the washing components of the plant. The pump shall be a close-coupled, self-priming centrifugal type pump mounted on a 4-wheel, pneumatic tired trailer capable of being towed by a 3/4-ton or larger truck over highways and unimproved roads. The pump shall be capable of supplying water from a source located at an elevation not less than 25 feet below the lowest part of the washing unit. The pump shall be of a size and capacity as recommended by the contractor to meet the requirements of the washing equipment. The pumping unit shall be furnished with all connections, accessories, hardware, safety equipment, engine and engine controls assembled complete for operation.

3.4.7 Water piping equipment. Water supply piping and water discharge piping or water flume shall be furnished complete with all valves, fittings, connectors, and hardware to assemble a complete water supply and discharge system. The system shall include but not be limited to the following components: A suction hose not less than 30 feet in length with terminal pipe and elbow for connecting to the pump, a foot valve to maintain pump prime, a suction strainer, pressure type water supply pipe, and discharge or waste water piping. The waste or discharge piping shall be of the circular corrugated (culvert) type with bolted flange joints and furnished in lengths of not less than 10 feet nor more than 20 feet. A total length of not less than 120 feet of discharge pipe shall be furnished. The pressure type water supply pipe shall be made up in sections not exceeding 20 feet in length. The sections shall be provided with quick-disconnect type gasketed joints. Not less than four sections shall be provided with bolt type flanges suitable for connecting to the pumping unit and to the water supply connector on the washing and screening unit. The total of not less than 200 feet of water supply pipe shall be furnished.

3.5 Semitrailers. Each unit of the construction plant and the construction support plant shall be separately mounted on a semitrailer chassis equipped with tandem axles and dual wheels. The semitrailers shall be capable of withstanding total loads imposed during highway and off-highway transport and during quarry operation. Semitrailers as required to transport auxiliary components of the plant shall be furnished with flat beds with tiedown provisions for securing the trailer load.

3.5.1 Suspension and axles. The axle suspension shall be the manufacturer's current commercial flexible, sprung, or articulated type. Axles and other load carrying components shall have capacity to support the maximum loads imposed under all operating conditions specified herein. Maximum load imposed on any axle for transport operation shall not exceed 25,000 pounds.

3.5.2 Wheels, tires, and tubes. The semitrailers shall be equipped with dual wheels. The tires shall be the tube or tubeless type with highway tread. All tires of a unit shall be the same size and have a rated capacity at least equal to the imposed load on the highest loaded tire with all equipment and components installed on the semitrailer. Tire flaps shall be provided if tube type tires are furnished.

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3.5.3 Vehicular electrical system. The semitrailer and dolly converter electrical system with accessories shall be furnished and installed in accordance with DOT Motor Carrier Safety Regulations.

3.5.4 Upper fifth wheel and kingpin. The front of the semitrailer shall be constructed to be compatible for coupling and towing by commercial truck tractor with anominal 54 inch fifth wheel height or a military equivalent truck tractor. The upper fifth wheel shall be of the full width type that permits coupling at any angle up to 90 degrees from the straight ahead position. The upper fifth wheel shall conform to SAE J701 and DOT Motor Carrier Safety Regulation 49 CFR 393.70.

3.5.5. Brake system. The semitrailer shall be equipped with a service airbrake system conforming to DOT Motor Carrier Safety Regulations and the Federal Motor Vehicle Standards. All components of the brake system shall be installed and protected to prevent damage from ground obstructions and debris and to obtain maximum ease of maintenance.

3.5.6 Dolly converter. A dolly converter shall be furnished with each semitrailer to convert from semitrailer to full trailer type operation for off-road transport and quarry type movement operations. The dolly shall be supported on a dual wheel sprung type, single or tandem axle system. The suspension system when installed shall be capable of withstanding towing and braking forces, off-road articulation, and shock loads under normal on-and-off-road operating conditions when coupled between the semitrailer and a truck, or other towing vehicle, without damage. The dolly shall be equipped with a heavy duty, fore and aft rocking type lower fifth wheel with oscillation lockouts and shall be compatible with the upper fifth wheel and kingpin furnished on the semitrailer front. The mounting of the fifth wheel shall be in accordance with DOT Federal Motor Carrier Service Regulations and shall be furnished with an A-frame type, hinged and lockable type tow mechanism suitable for off-road use. A towbar eye in accordance with SAE J847 shall be furnished and shall be installed in accordance with SAE J849. The dolly shall be provided with an airbrake system as specified in 3.5.5. In addition, the dolly shall be equipped with all intervehicular airbrake and electrical lines necessary to connect to the airbrake and electrical systems of the towing vehicle to the semitrailer systems. All connections shall be of sufficient length and installed to allow the dolly to turn at least 110 degrees to the right and left from the straight ahead position without damage.

3.6 Operator stations. The construction plant and the construction support plant shall each be capable of being independently operated by one man from a remote control operator station. The stations shall be accessible to operator, protected from falling debris, moving machinery, and components, and shall be located in such a manner to provide the operator with an unobstructed view of the over-all plant operation. The operator station shall be equipped with push-button type remote electrical controls which will enable the operator to control each individual operating component of the plant. The controls shall be of a type and installed in such a manner to provide for sequential starting of the plant components to protect against electrical overload of power source and to maintain a continuous, uninterrupted flow of materials through the plant. Each control in the station shall be individually marked and identified as to its component function. Complete starting, stopping, and other operational instructions as required for safe and proper operation of the plant shall be printed on metal plates, mounted and located clearly visible to the operator in the operator station.

3.7 Power distribution equipment. The construction plant and the construction support plant shall each be equipped with a power distribution and control cabinet. The cabinets shall be capable of being combined for the construction support plant operation or connected directly to the power source for individual plant type operation. Each cabinet shall also be capable of being connected directly to a commercial power source or to a portable generator type power source. The cabinets shall be installed on the plants and furnished complete with all required overload protected motor starters, circuit breakers, busses and internal wiring and terminal connections. The motor starters shall be the across-line magnetic type or the reduced voltage type as recommended for the horsepower requirement of the individual electric motor. The cabinets shall be rated and bussed for 240/416 volt, 3-phase, 60 cycle electric current. External motor leads shall be mine cable type wiring with lug type connectors at the motors and quick-disconnect

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watertight type couplings at the power distribution cabinet. All interconnecting wiring between the distribution cabinets, controls stations, and motors shall be furnished. All electrical equipment shall be positively grounded to assure an electrical bond between the equipment and the unit on which is is mounted. All electrical equipment as furnished and installed shall conform to the National Electrical Manufacturers Association Standards as applicable.

3.8 Lubrication. The crushing, screening, and washing plant shall be capable of operation with military standard lubricants without deleterious effects due to lubrication.

3.9 Lifting and tiedown. Lifting and tiedown attachments, if available either as standard or optional equipment by the manufacturer shall be furnished. If the attachments are not offered or available, suitable instructional procedures for lifting and tying down the trailer units and major components shall be provided. The instructions shall be provided on corrosion-resistant plates and permanently affixed on each semitrailer unit in a conspicuous location.

3.10 Painting. The crushing plants and auxiliary equipment shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice. The final color shall be forest green in accordance with MIL-E-52798.

3.11 Noise. If the noise level at the operator's station exceeds 85dB(A) as measured in accordance with SAE J919, a warning plate shall be affixed at the operator's station in a conspicuous and protected location. The sign shall conform to Figure 1 and shall be visable to all personnel exposed to the hazard.

3.12 Safety. The crushing plants shall comply with the applicable safety requirements of the OSHA standards which are in effect on the date of the opening of the bids.

3.13 Identification plates and marking.

(a) All components of each plant shall be provided with a plate made of noncorrosive metal, 0.035 inch minimum thickness, and stamped with letters not less than 0.25 inch high, containing the following information:

Nomenclature
NSN Number
Serial Number
Model
Contract Number
Shipping date
USA Number
Shipping Weight
Shipping Cubage

(b) The lettering "US ARMY" and the US Army Registration Number shall be applied to each semitrailer. The lettering shall be 3 inches high and the color shall be lusterless black 37038 in accordance with FED. STD. No. 595. Each auxiliary component of the crushing, screening and washing plant shall be provided with a plate made of materials as specified above containing the following information:

Nomenclature
NSN
Manufacturer
Model
For Use With

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3.14 Special tools. Any special tools required for operation or operator maintenance of the crushing, screening and washing plant shall be furnished by the manufacturer with each plant.

3.15 Toolbox. Each semitrailer unit of the crushing plant shall be provided with a lockable type toolbox or closed compartment for storage of tools and accessories required for operation.

3.16 Government-furnished equipment (GFE). The following will be furnished by the Government for use under the contract:

(a) Case, Maintenance and Operator's Manual, NSN 7520-00-559-9618 complete with log book, binder and forms.
(b) User Maintenance Support Plan.

3.17 Mounting. The manufacturer shall furnish all fasteners and install on each crushing plant the GFE maintenance and operator's manual case as specified in 3.16(a). The case shall be installed at the operator control station accessible to the operator.

3.18 Electric power source. Unless otherwise specified (see 6.1), the Government will furnish two 500 kw, 240/416 volt, 60 Hz portable generator sets to be utilized as a power source for the electric motors as furnished to power the crushing, screening and washing plant and auxiliary equipment. When specified (see 6.1), the contractor shall submit with the bid or technical proposal the total electrical horsepower requirements of his crushing, screening and washing plant as proposed to meet the production requirements as specified herein.

3.19 Transportability data. The Government will collect and record transportability data. The contractor shall make the crushing plants available to the Government for the required determinations.

3.20 Technical publications. Technical publications shall be furnished as specified in the contract or purchase order (see 6.1).

3.21 Certifications. The manufacturer shall provide within the specified time frame (see 6.1) written certification that the crushing, screening and washing plant with all components, auxiliary equipment and materials as furnished conform as applicable to the requirements of the following to the extent specified herein:

(a) Department of Labor (OSHA) Regulations.
(b) Department of Transportation (MCSR) Federal Motor Carrier Safety Regulations.
(c) Department of Transportation Federal Motor Vehicle Standards.
(d) Required Society of Automotive Engineers Standards.
(e) Suitability for operation in the ambient temperature range as specified in 3.2.
(f) Suitable for transport as semitrailer units truck tractor or the military equivalent truck tractor.
(g) That the crushing, screening and washing plant will meet all production requirements as specified herein.

3.22 Technical Information Package (TIP). The information in the TIP shall be completed depicting the crushing, screening and washing plant and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year, and shall include identification of the component or part previously used. The manufacturer shall certify, by signing the TIP, that the item to be furnished is identical to those being procured for commercial use, except for modifications permitted in 3.1 of the specification. The TIP shall be submitted with all bids and proposals.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article evaluation and demonstration. Prior to delivery under the contract, the manufacturer shall make a crushing, screening and washing plant available for first article evaluation and demonstration. The plant and its components shall be compared with this document, the manufacturer's completed Technical Information Package (see 3.22), and published literature, calculation and test data, to verify compliance with this specification. The plant shall then be subjected to a demonstration and operational test as specified herein. Permanent deformation, overheating, malfunction, failure to meet performance requirements, failure to complete any test, or leakage of lubricants shall be basis for rejection of the first article plant. If such deficiencies do arise, they shall be corrected and the first article evaluation and demonstration shall be repeated.

4.2.1 Assembly and examination. The crushing, screening and washing plant shall be assembled with components in the configuration as specified in the contract or purchase order and made available by the manufacturer to the Government for examination. The plant when assembled will include all functioning components, each located and attached in its respective operating position to produce crushed rock and aggregate. The plant and components shall be compared with the Technical Information Package (TIP) (see 3.22) as submitted by the manufacturer to verify conformance.

4.2.2 Servicing plant. The plant and all components and accessory equipment shall be lubricated as recommended by the manufacturer with military type lubricants. All component and accessory drive belts, conveyor belts, crushing and screening units shall be adjusted for operation. Install and connect electrically all electric drive motors, electrical control equipment wiring and operator control stations. All walkways, safety guards, access ladders and other equipment as required to complete the plant shall be installed, secured and adjusted for operation to simulate the actual production of concrete and asphalt aggregate.

4.2.3 Operational test. Unless otherwise specified, the two 500 kw military generator plants as furnished by the Government (see 3.18) shall be connected to the power distribution cabinets of the crushing, screening and washing plant. When specified (see 6.1), the power distribution cabinets may be temporarily connected to available commercial power as a power source. Operate the completely assembled plant including all components and accessories for a period of 1 hour without crushing rock or producing aggregate. Any operational difficulties, as specified in 4.2, shall be cause for rejection.

5. PACKAGING.

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.1), each complete plant and all components and accessories with the plant shall be preserved and packed for shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings in compliance with carrier regulations applicable to the mode of transportation. In addition, the following requirements shall be accomplished. Items with critical surfaces required for fit and function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors of desiccation packs. Items that require protection from physical damage shall be wrapped or boxed. Instructions shall be furnished indicating the preservation or packing materials to be removed and any other servicing required prior to placing the plant in operation. Marking of the components shall reflect as a minimum the item name, stock number, quantity, contract or purchase order number and appropriate address markings on each unpacked item

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and shipping container. The marking shall be applied with a waterproof material and shall provide a definite contrast.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When commercial electric power may be used as a power source (see 3.18 and 4.2.3).
(c) When GFE generators will be used as a power source (see 3.18).
(d) Time frame and type of technical publications to be furnished (see 3.20).
(e) Time frame for submitting certifications (see 3.21).
(f) When preservation, packing, and marking shall be other than specified (see 5.1).
(g) Time frame for completing and submitting a Technical Information Package (see 3.22).

6.2 Government furnished property. The contracting officer should arrange to furnish the contractor the items specified in 3.9.13.

6.3 Publications. The contracting officer shall determine that the manuals (operator, parts, maintenance, and overhaul) and the lubrication instructions proposed to be furnished by the contractor are approved by the Government prior to accepting delivery of the item.

Custodian:
Army – ME

Preparing activity:
Army – ME

Review activity:
Army – AT

Project No. 3820-F159

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FIGURE 1. Noise hazard caution sign. X-3181

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52121	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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